

Employee Leave Management System - MySQL Capstone Project

Project Objective: Design and implement an Employee Leave Management System using MySQL to manage employees, departments, leave types and leave requests. Apply normalization, constraints, CRUD, joins, numeric & string & date functions, GROUP BY, HAVING, subqueries, views and transactions.

Problem Statement

Organizations need a centralized system to manage employee leave requests, approvals and reporting. Build a relational database that stores leave information and supports operational and analytical queries.

MySQL Workflow

- Requirement Analysis
- Identify entities (Departments, Employees, Managers, Leave Types, Leave Requests)
- Draw ER Diagram
- Normalize to 3NF
- Create database and tables
- Insert sample data
- Perform CRUD
- Use String, Numeric & Date Functions
- Write joins and subqueries
- Create views
- Perform transactions
- Generate reports

Database Design

Departments(department_id PK, department_name)

Employees(employee_id PK, department_id FK, employee_name, phone, email)

Managers(manager_id PK, manager_name)

Leave_Types(leave_type_id PK, leave_type_name)

Leave_Requests(leave_id PK, employee_id FK, manager_id FK, leave_type_id FK, start_date, end_date, total_days, status, reason)

Sample Data

-- Departments

INSERT INTO Departments VALUES

(1,'HR'), (2,'IT'), (3,'Finance'), (4,'Sales'), (5,'Administration');

-- Managers

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INSERT INTO Managers VALUES
(1,'Ramesh'),(2,'Priya'), (3,'Kiran'), (4,'Sneha'), (5,'Arun');
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-- Leave Types
INSERT INTO Leave_Types VALUES
(1,'Casual Leave'),
(2,'Sick Leave'),
(3,'Earned Leave'),
(4,'Maternity Leave'),
(5,'Unpaid Leave');
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-- Employees
INSERT INTO Employees VALUES
(101,1,'Aarav','9876500001','aarav@mail.com'),
(102,2,'Bhavya','9876500002','bhavya@mail.com'),
(103,3,'Charan','9876500003','charan@mail.com'),
(104,4,'Divya','9876500004','divya@mail.com'),
(105,5,'Esha','9876500005','esha@mail.com'),
(106,2,'Farhan','9876500006','farhan@mail.com'),
(107,1,'Gopi','9876500007','gopi@mail.com'),
(108,3,'Hari','9876500008','hari@mail.com'),
(109,4,'Isha','9876500009','isha@mail.com'),
(110,5,'John','9876500010','john@mail.com');
```

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-- Leave Requests
INSERT INTO Leave_Requests VALUES
(1001,101,1,1,'2026-07-01','2026-07-03',3,'Approved','Personal work'),
(1002,102,2,2,'2026-07-02','2026-07-04',3,'Pending','Fever'),
(1003,103,3,3,'2026-07-05','2026-07-06',2,'Approved','Vacation'),
(1004,104,4,1,'2026-07-07','2026-07-08',2,'Rejected','Family function'),
(1005,105,5,5,'2026-07-10','2026-07-12',3,'Approved','Personal'),
(1006,106,2,2,'2026-07-11','2026-07-11',1,'Pending','Medical'),
(1007,107,1,1,'2026-07-13','2026-07-15',3,'Approved','Travel'),
(1008,108,3,3,'2026-07-16','2026-07-20',5,'Approved','Vacation'),
(1009,109,4,2,'2026-07-18','2026-07-19',2,'Rejected','Sick'),
(1010,110,5,1,'2026-07-21','2026-07-22',2,'Pending','Personal');
```

SQL Questions

1. List all leave requests.
2. Display leave request with employee name.
3. Show approved leave requests.

4. Show pending leave requests.
5. Employees currently on leave.
6. Count leave requests department-wise.
7. Count leave requests by status.
8. Average leave days by department.
9. Employee with maximum leave days.
10. Department with maximum leave requests.
11. Employees with more than 2 leave requests.
12. Employees who never applied for leave.
13. Latest 5 leave requests.
14. Leave requests in last 30 days.
15. Display employee names in uppercase and lowercase.
16. Show first 3 characters of employee names.
17. Calculate leave duration.
18. Generate Leave Reference (EL-1001).
19. Leave types used more than 5 times.
20. Manager-wise approved leave count.
21. Employees whose leave exceeds department average.
22. Show monthly leave statistics.
23. Longest leave taken.
24. Remaining leave balance.
25. Open leave requests.
26. Rejected leave requests.
27. Create a view for approved leaves.
28. Create a view for pending leaves.
29. Transaction to approve leave request.

30. Transaction to cancel leave request and restore balance.

31. Create a stored procedure to display all leave requests handled by a particular manager.

- Input: manager_id
- Display employee name, leave type, start date, end date, and status.

32. Create a stored procedure to calculate the total number of leave days taken by an employee.

- Input: employee_id
- Consider only leave requests with status Approved

33. Create a trigger on the Leave_Requests table.

- Whenever a new leave request is inserted, automatically store the leave_id, employee_id, leave_type_id, and status in a Leave_Request_Log table.
- The log table should also store the date on which the request was created.

34. Create a trigger on Leave_Requests.

- Whenever a leave request's status changes, store the leave_id, employee ID, old status, new status, and change date in a Leave_Status_History table.